Department of Economics
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UN3412
Spring 2026

Problem Set 6
Introduction to Econometrics
for Section 1
(Erden)

This question deals with the dataset of American Express credit card applicants **AmEx.dta**. To be able to view the data set with STATA, you may need to write "set memory 32m" on the command window first. For each part, you will need to report results for the linear probability model, logit, and probit models. The best way to organize your results is in the form of a table. Report the log-likelihood and pseudo R^2 for the nonlinear models and the R^2 and adjusted R^2 for the linear model

cardhldr: a binary indicator for whether or not the individual is approved as cardholder
income the applicants income
age: the applicants age in years
selfempl: a binary indicator for whether or not the individual is self-employed
ownrent: an indicator for whether or not an individual owns or rents their primary residence
acadmos: the number of months and individual has resided at their current address
majdrg: the number of major derogatory credit events in the last 12 months
mindrg: the number of minor derogatory credit events in the last 12 months.

(a) (3p) First, To get the scale of the coefficients right, **`replace`** $\boldsymbol{income} = \boldsymbol{income/1000}$. You only need a few of the variables from the data set. So, it is a good idea to keep those and not the rest. You can do this with the following command: **`keep cardhldr income age selfempl ownrent acadmos majordrg minordrg`**
Now, what fraction of consumers in the sample are cardholders? What does this tell us in terms of external validity? (*hint:* use **`tabulate`** command)

(b) (3p) Consider predicting whether or not an individual is approved as a cardholder *cardhldr* as a function of *income, age, selfempl, ownrent, acadmos*. Write commands for LPM, logit and probit.

(c) (8p) Compute the marginal effect (how does $Pr(Y = 1)$ change) for someone who is self-employed and for whether or not they rent or own their home. Evaluate this for an individual who has median values for their X variables (covariates).

	LPM	$Logit$	$Probit$
$\Delta selfemp$			
$\Delta ownrent$			
$\%\Delta selfemp$			
$\%\Delta ownrent$			

(d) (15p) Repeat the previous exercise for someone whose income is at the 20% and 80% of the consumers in the sample.

	LPM	$Logit$	$Probit$
$\Delta selfemp$			
$\Delta ownrent$			
$\%\Delta selfemp$			
$\%\Delta ownrent$			

	LPM	$Logit$	$Probit$
$\Delta selfemp$			
$\Delta ownrent$			
$\%\Delta selfemp$			
$\%\Delta ownrent$			

(e) (5p) Express all three cases (20%, 50%, 80%) in terms of the probability ratio or log-odds. How big is the impact? Is this a lot or a little?

(f) (4p) Now consider a specification that includes major and minor derogatory credit events in the last year (*majdrg,mindrg*). Write commands for LPM, logit and probit.

(g) (18p) Report the full table of specifications (there should be 6) with all relevant statistics. Mark 1%, 5% and 10% significance levels using the key at the bottom of the table.

$VARIABLES$	(1) LPM	(2) $Logit$	(3) $Probit$	(4) LPM	(5) $Logit$	(6) $Probit$
$income$	( )	( )	( )	( )	( )	( )
age	( )	( )	( )	( )	( )	( )
$selfemp$	( )	( )	( )	( )	( )	( )
$ownrent$	( )	( )	( )	( )	( )	( )
$acadmos$	( )	( )	( )	( )	( )	( )
$majordrg$	–	–	–	( )	( )	( )
$minordgr$	–	–	–	( )	( )	( )
$Constant$	( )	( )	( )	( )	( )	( )
$Observations$						
R^2		–	–		–	–
$Pseudo\ R^2$	–			–		

$Robust\ standard\ errors\ in\ parentheses$

$^{**}\ p < 0.01,\qquad ^{*}\ p < 0.05,\qquad ^{+}\ p < 0.1$

(h) (6p) Which model fits the data the best? Which variables have the most explanatory power? Do major derogatory credit events matter? Do you prefer models with or without derogatory events?

(i) (6p) Construct a 95% confidence interval for the effect of an additional major derogatory credit event has on being an American Express cardholder under the linear probability model.

(j) (6p) Repeat for the logit and probit models in the case where the x values take on the median values in the sample.

(k) (8p) Compute the fitted values for the linear probability model, and plot a histogram for the fitted values of the y variable. Do the same for logit and probit models.

(l) (6p) What should be the domain of predicted values from the LPM? What fraction of values lies outside this domain? How do we interpret them?

(m)(6p) From the results you find on Table 1, what can you say about probit and logit results? How about LPM results?

(n) (6p) A well-known result by Amemiya (1981) is that over the range of probabilities from 30% to 70% the following approximations should hold: $\beta_{LPM} \approx 0.4\ \beta_{probit}$ and $\beta_{LPM} \approx 0.25\ \beta_{logit}$. Take your results for logit and probit, how does this result look for the $\beta_{majordrg}$?

<u>Following questions will not be graded, they are for you to practice and will be discussed at the recitation:</u>

1. [ungraded] SW Empirical Exercise 11.1

2. [ungraded] SW Empirical Exercise 11.2